higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

21 NOVEMBER 2019

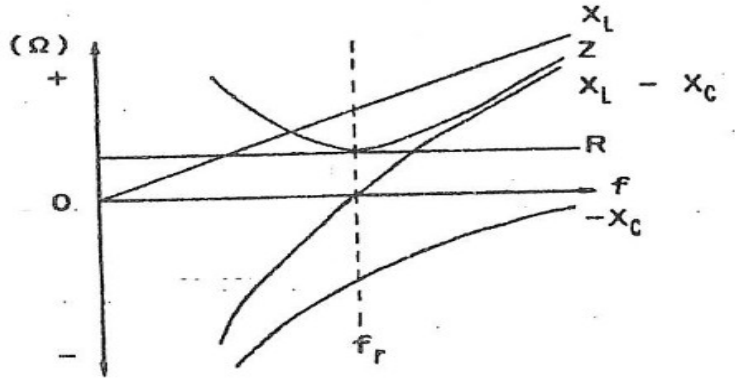
This marking guideline consists of 10 pages.

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QUESTION 1

1.1 They block the DC signal from the reference of the AC signal. (2)

1.2



REACTANCE, RESISTANCE AND IMPEDANCE AS OPPOSED TO FREQUENCY (4)

1.3 1.3.1 $Xc = \frac{1}{2\pi fC}$

$= \frac{1}{2 \ x \ \pi \ x \ 100 \ x \ 200 \ x \ 10^{-6}}$

$= 15{,}915 \ \Omega$

$Z_A = 15 - j \ 15{,}915 \ \Omega = 21{,}87 < -46{,}7^0 \ \Omega$

$I_A = \frac{V_T}{Z_A}$

$= \frac{200 < 0^0}{21{,}87 < -46{,}7^0}$

$= 9{,}145 < 46{,}7^0 \ A$

$I_A = 6{,}272 + j \ 6{,}656 \ A$

$I_T = I_A + I_B$

$\therefore I_B = I_T - I_A$

$= 15 + j \ 0 - (6{,}272 + j \ 6{,}656)$

$= 8{,}728 - j \ 6{,}656$

$= 10{,}976 < -37{,}33^0$ ⟶ (7)

1.3.2 $Z_B = \frac{V_T}{I_B}$

$= \frac{200 \angle 0^0}{10,976 \angle -37,33^0}$

$= 18,222 \angle 37,33^0$ ⟶ (2)

1.3.3 $Z_B = 14,49 + j\ 11,05\ \Omega$

$R = 14,49\ \Omega$ & $X_L = 11,05\ \Omega$ ⟶ (1)

[16]

QUESTION 2

2.1 2.1.1 $r' = \frac{V'_{r(rms)}}{V'_{DC}}$

$\therefore V'_{DC} = \frac{V'_{r(rms)}}{r'}$

$= \frac{0,118}{0,013}$

$= 9,08\ V$

$V'_{DC} = \frac{R_L}{R_L + R} x\ V_{DC}$

$\therefore V_{DC} = \frac{(R_L + R)}{R_L}\ x\ V'_{DC}$

$= \frac{(1\ 200 + 450)}{1\ 200} x\ 9,08$

$= 12,5\ V$ ⟶ (4)

2.1.2 $V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} x\ V_{r(rms)}$

$\therefore V_{r(rms)} = \frac{V'_{r(rms)}\sqrt{R^2 + X_C^2}}{X_C}$

$= \frac{0.118\ x\ \sqrt{450^2 + 26,526^2}}{26,526}$ (3)

$$= \quad 2{,}005\ V \longrightarrow$$

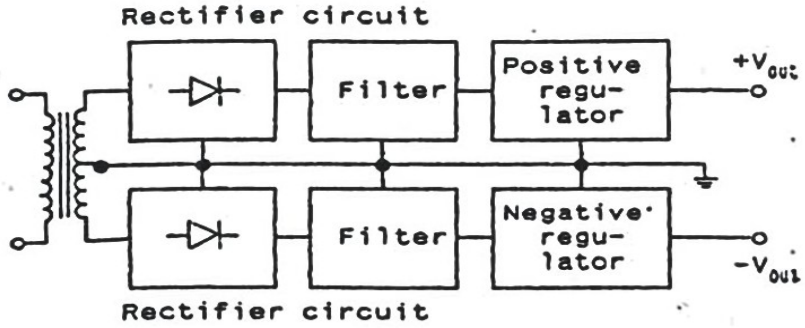
2.1.3 $$r = \frac{V_{r(rms)}}{V_{DC}}$$

$$= \frac{2{,}005}{12{,}5}$$

$$= \quad 0{,}161 \longrightarrow \quad (2)$$

2.2 It is the amount of change the output DC-voltage experience during varying load conditions. (2)

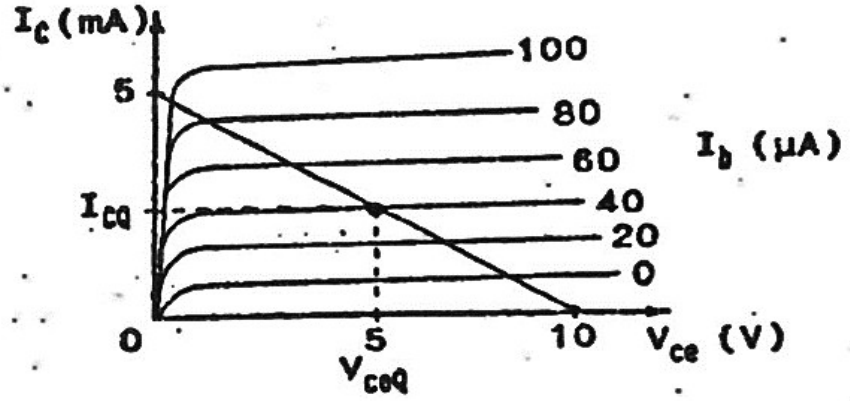
2.3



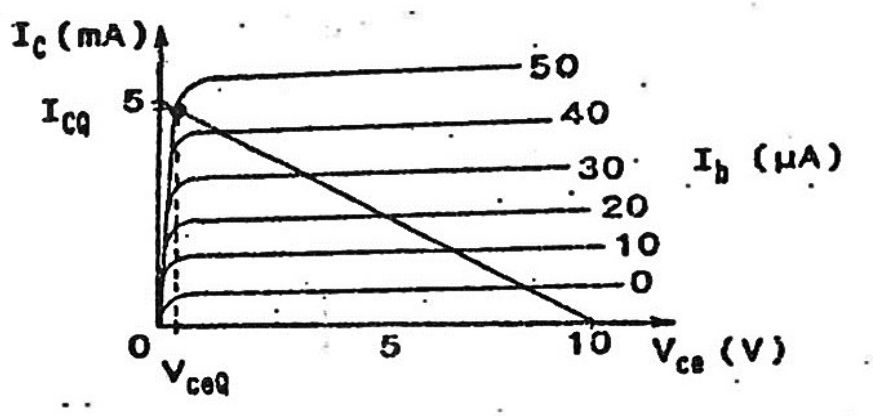
DOUBLE POWER SUPPLY (6)
[17]

QUESTION 3

3.1 3.1.1



OPERATING POINT AT A NORMAL TEMPERATURE (25 °C) (2)



OPERATING POINT AT A HIGH TEMPERATURE (100 °C) (2)

3.1.2 To prevent the shifting of the operating point, an emitter resistor, R_e, is connected between the emitter and ground. The function of R_e is to limit the collector current during the increase in temperature. (3)

3.2 $R_{B2} = \frac{1}{10}\ \beta R_e$

$\therefore R_e = \frac{10\,R_{B2}}{\beta}$

$= \frac{10\ x\ 7\ 400}{200}$

$= 370\ \Omega$

$C_e = \frac{10}{2\pi f\ R_e}$

(4)

$$= \frac{10}{2 \ x \ \pi \ x \ 50 \ x \ 370}$$

$$= 86{,}03 \ \mu f \longrightarrow$$

3.3 3.3.1

$$Z_1 = h_{ie} - \frac{h_{fe} . h_{re} . Z_L}{1 + h_{oe} . Z_L}$$

$$= 1\ 000 - \frac{50 \ x \ 2{,}5 \ x \ 10^{-4} \ x \ 2\ 200}{1 + 25 \ x \ 10^{-6} \ x \ 2\ 200} \quad \surd$$

$$= 973{,}934$$

$$Z_i = \frac{R_b x \ Z_1}{R_b + Z_1}$$

$$= \frac{200 \ x \ 10^3 \ x \ 973{,}934}{200 \ x \ 10^3 + 973{,}934} \quad \surd$$

$$= 969.214 \ \Omega \longrightarrow \quad (3)$$

3.3.2

$$A_V = \frac{-h_{fe} . Z_L}{h_{ie} + \left(h_{ie} . h_{oe} + h_{fe} h_{Re}\right) Z_L}$$

$$= \frac{-50 \ x \ 2\ 200}{1\ 000 + \left(1\ 000 \ x \ 25 \ x \ 10^{-6} - 50 \ x \ 2{,}5 \ x \right.}$$

(2)

$$= 110 \longrightarrow$$

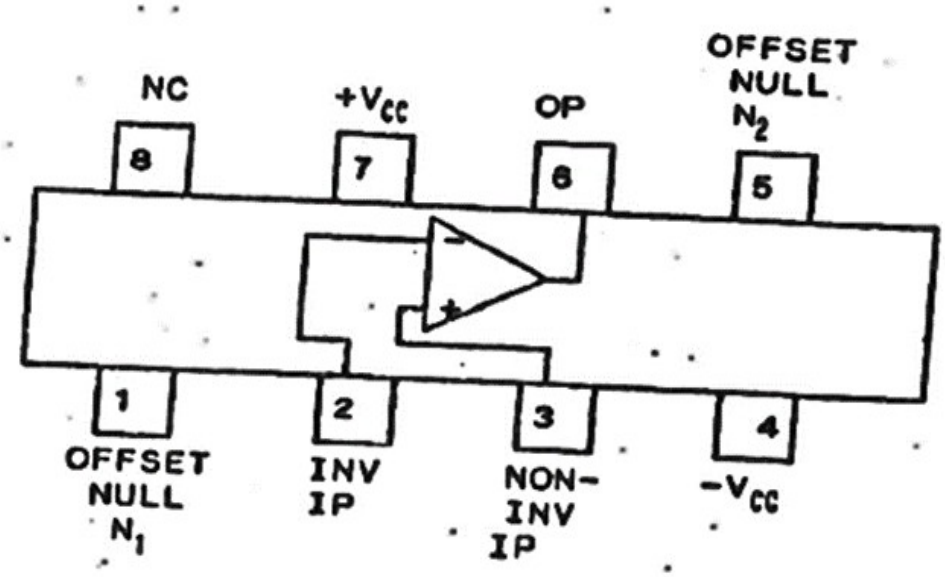
[16]

QUESTION 4

4.1
- Drift
- Noise

(2)

4.2



(4)

741-OPERATIONAL AMPLIFIER IN A DUAL IN-LINE PACKAGE

4.3 $t = \frac{1}{f}$

$= \frac{1}{2\ x\ 10^{3}}$

$= 0{,}5\ ms$

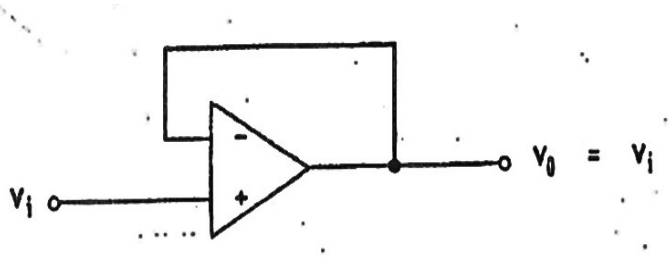
$t = R_f . C$

$\therefore R_f = \frac{t}{C}$

$= \frac{0{,}5\ x\ 10^{-3}}{2\ x\ 10^{-6}}$

$= 250\ \Omega \longrightarrow$ (3)

4.4

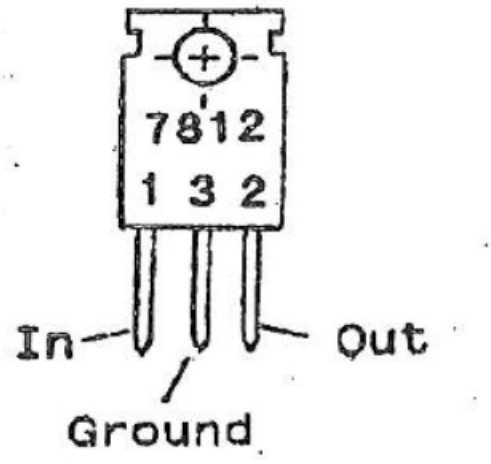


THE VOLTAGE FOLLOWER (2)

[11]

QUESTION 5

5.1

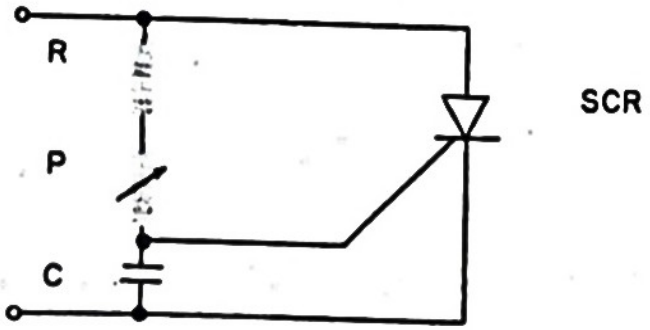


CONNECTION OF 7812 TERMINAL (3)

5.2 All unutilised inputs must be connected to V_{DD} or V_{SS}. (2)
[5]

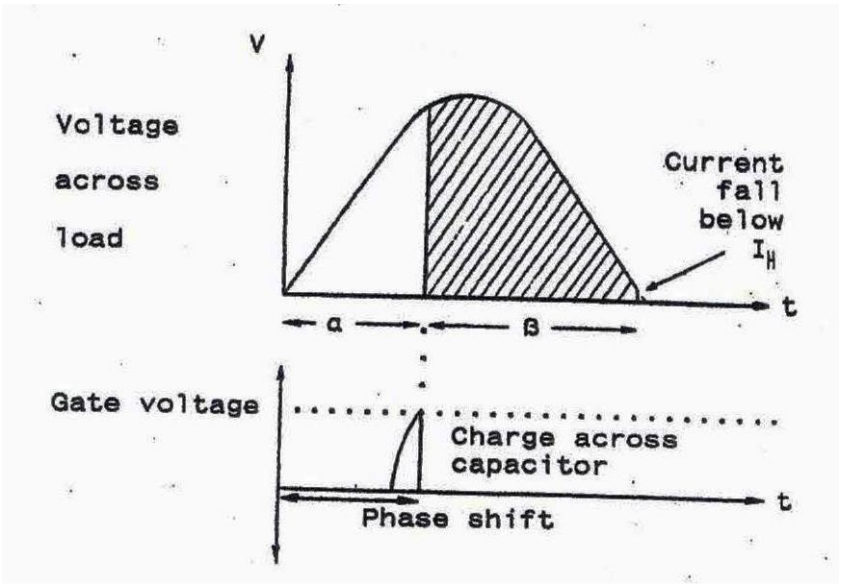
QUESTION 6

6.1



TRIGGER CIRCUIT (2)

6.2



SINGLE-PHASE CONTROLLED CIRCUIT

(4)
[6]

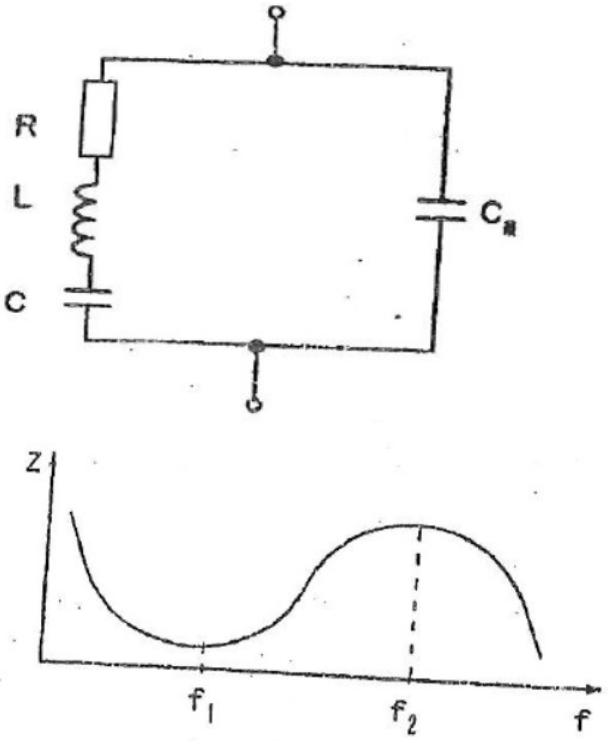
QUESTION 7

7.1 Analog
7.2 Unknown
7.3 Comparator
7.4 Code
7.5 Equivalent
7.6 Compared

(6 × 1) **[6]**

QUESTION 8

8.1

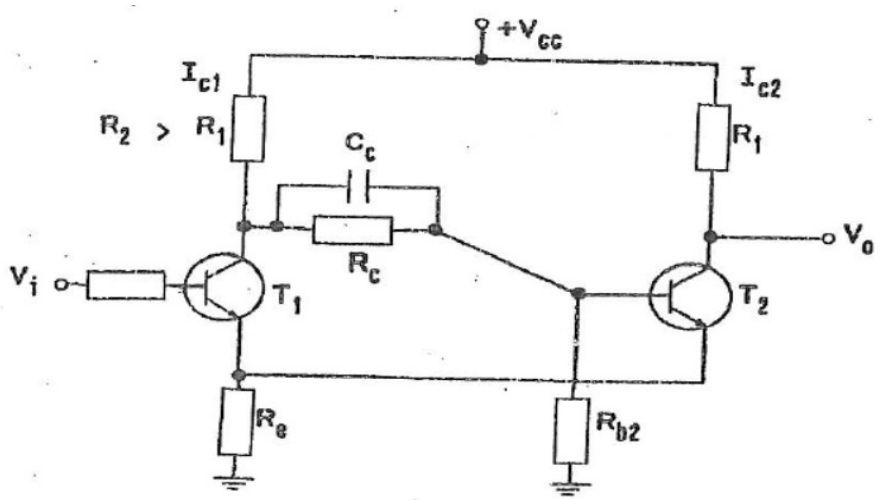


(3)

- Series resonance takes place at a certain frequency f_r when $jX_L = -jX_C$ (C_M is excluded), Z = R and is the minimum.
- Parallel resonance takes place at a higher frequency f_2 when $R + j(X_L - jX_C) = -jX_C$, Z = R and is the maximum. (2)

8.2 The Rozner modification is a method employed to increase the rise time at the collector of a transistor, T_1. (2)

8.3



THE SCHMITT TRIGGER CIRCUIT (4)

[11]

QUESTION 9

9.1 $Resolution = \frac{1}{No.\ of\ turns}$

$= \frac{1}{350}$

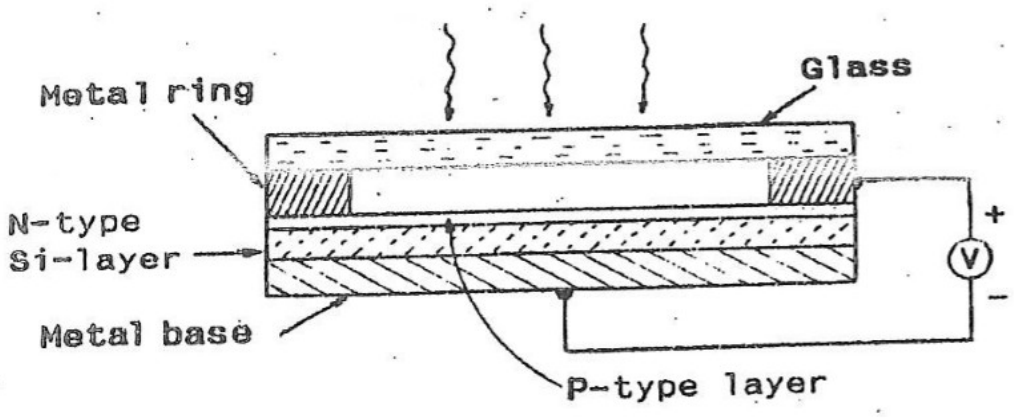
$= 2{,}86 \times 10^{-3}$ ⟶ (2)

9.2

9.2.1	Indorsed
9.2.2	External
9.2.3	Magnetic flux
9.2.4	Induced
9.2.5	Greater
9.2.6	Size
9.2.7	Difference
9.2.8	Secondary
9.2.9	Displacement
9.2.10	Phase

(10 × ½) (5)

9.3



THE PHOTOVOLTAIC CELL (5)

[12]

TOTAL: 100

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